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DETAIL SPECIFICATION

LABELS, GARMENT (WOVEN, RAYON)

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements of a woven, rayon garment labels for clothing items.

1.2 Classification. This specification covers the following types and sizes.

1.2.1 Types.

Type I – Rayon
Type II - Polyester

1.2.2 Sizes.

Size A – 1 5/16-inches wide by 3-inches long
Size B – 3/4-inch wide by 2-inches long
Size C – 2 1/4-inches wide by 4-inches long
Size D – 2 1/4-inches wide by 5-inches long

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. N/A

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of these documents are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1	- Evaluation Procedure for Gray Scale for Color Change
AATCC EP2	- Evaluation Procedure for Gray Scale for Staining
AATCC EP9	- Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM20	- Test Method for Fiber Analysis: Qualitative
AATCC TM15	- Test Method for Colorfastness to Perspiration
AATCC TM117	- Test Method for Colorfastness to Heat: Dry (Excluding Pressing)
AATCC TM132	- Test Method for Colorfastness to Drycleaning

(Copies of these documents are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

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(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D1907/D1907M - Standard Test Method for Linear Density of Yarn (Yarn Number) by Skein Method
ASTM D3775 - Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics

(Copies of these documents are available online at https://www.astm.org.)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies of this document are available online at https://www.crcpress.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Not applicable

3.2 Standard sample. Unless otherwise indicated, the finished labels shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials.

3.4.1 Type I. The type I labels shall be woven using filament rayon of the regenerated cellulose type (viscose or cuprammonium) in the denier specified in Table I.

3.4.2 Type II. The type II labels shall be woven using filament polyester (polyethylene glycol terephthalate) in the denier specified in Table II.

3.4.3 Weave. The labels shall be woven in accordance with Table I and II in a taffeta and figured weave, when tested as specified in 4.5.

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TABLE I. Label weave construction - Type I

Characteristic	Yarn size	Count
Ground warp	100 denier singles or 50 denier, 2 ply	144 ends per inch (min.)
Ground filling	75 denier singles or 100 denier singles	92 picks per inch (min.) 84 picks per inch (min.)
Figure filling	150 denier singles	84 picks per inch (min.)
Selvage (size A and B)	100 denier singles or 50 denier, 2 ply or 150 denier singles	8 double ends each selvage 8 single ends each selvage
Selvage (size C and D)	100 denier singles or 50 denier, 2 ply	12 double ends each selvage

TABLE II. Label weave construction - Type II

Characteristic	Yarn size	Count
Ground warp	90 denier singles or 40 denier, 2 ply	144 ends per inch (min.)
Ground filling	70 denier singles or 40 denier, 2 ply	88 picks per inch (min.)
Figure filling	150 denier singles	88 picks per inch (min.)
Selvage (size A and B)	90 denier singles or 40 denier, 2 ply	8 double ends each selvage
Selvage (size C and D)	90 denier singles or 40 denier, 2 ply	12 double ends each selvage

3.5 Color. The label background and selvage shall be black 211 and the design and legends shall be white, or as specified in the contract or purchase order.

3.5.1 Visual shade matching. The color and appearance of the label shall match the standard sample when tested as specified in 4.6.1.

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3.6 Labile sulfur (Type I only). Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used. The dyed yarn/label shall be "Free" of labile sulfur when tested as specified in 4.6.2.

3.7 Colorfastness. Unless otherwise specified in the contract or procurement documents, the finished label referenced in 3.4 shall conform to the colorfastness requirements listed below in Table III, when tested as specified in 4.5.

TABLE III. Colorfastness requirements (all types).

Color Evaluation	Perspiration (Acid & Alkaline) (Color Change and Staining) (min.)	Dry Cleaning (Color Change and Staining) (1 cycle) (min.)	Heat – Dry (Type II only) (min.)
All colors	4	4	4

3.8 Size. The size shall be as specified (see 1.2.2 and 6.2) with a tolerance of minus 1/16-inch and plus 3/16-inch in length and width.

3.9 Design and legend. The design and the legend of the labels including the size characters shall be in accordance with end item specification requirements or as specified in the contract or purchase order (see 6.2).

3.10 Finish. The finish of the label shall be equal to or better than that of the standard sample for appearance and hand.

3.11 Toxicity. The finished label shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.6.3. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.12 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.4 and 4.5, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. Not applicable

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4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.6 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurements documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual examination. The labels shall be examined for the defects listed in Table IV. The lot size shall be expressed in units of labels. The sample unit shall be one (1) label.

TABLE IV. End item visual defects.

Examine	Defect description	Classification	
		Major	Minor
Fabric	- Cut, hole or tear	101	
	- Broken ends	102	
	- Tight end 1/	103	
	- Slack end forming loops on surface 1/	104	
Labels	- Misshapened or distorted	105	
	- Design or legend incomplete, illegible, incorrect or distorted	106	
	- Size characters not as specified, incorrect	107	
	- Selvage cut, torn, broken, loopy, stringy or tight to an extent causing puckering	108	
	- Hook, needle or card mark – affecting legibility	109	
	- Fold mark, where required, omitted or misplaced		201
	- Hard crease		202
Color	- Not specified color for background, selvage, design, and legends	110	
Finish	- Does not match standard sample for appearance and hand		203
Cleanness	- Spot, stain or discoloration 1/	111	
	- Objectionable odor	112	

1/ Clearly visible at normal inspection distance.

4.5 End item testing. The labels shall be tested for the characteristics listed in Table V. The methods of testing as specified wherever applicable and as listed in Table V shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be eight (8) finished labels for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

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Lot size (label)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE V. Material/label physical and chemical testing.

Characteristics	Types	Requirement Reference	Test method
Fiber identification	I, II	3.4.1 and 3.4.2	AATCC TM20 (see 6.4)
Weave	I, II	3.4.3	Visual 1/
Yarn Denier	I, II	Table I, II	ASTM D1907/D1907M
Yarns per inch	I, II	Table I, II	ASTM D3775
Visual shade matching	I, II	3.5	AATCC EP9, Option C and 4.6.1
Labile sulfur	I	3.6	4.6.2
Colorfastness: Perspiration (Acid & Alkaline) Dry Cleaning Heat – Dry	 I, II I, II II	 Table III Table III Table III	 AATCC TM15 2/ AATCC TM132 3/ AATCC TM117 2/, 4/
Size	I, II	3.8	Calibrated ruler
Toxicity	I, II	3.11	4.6.3

1/ One determination shall be made per sample unit and results reported as “pass” or “fail”.
2/ Rated using the AATCC EP1, Evaluation Procedure for Gray Scale for Color Change and AATCC EP2, Evaluation Procedure for Gray Scale for Staining.
3/ Rated using the AATCC EP1, Evaluation Procedure for Gray Scale for Color Change.
4/ The test shall be conducted at 150 (± 2)°C.

4.6 Methods of testing and inspection.

4.6.1 Visual shade matching. The color and appearance of the cloths shall match the standard sample when viewed using the AATCC EP9, Evaluation Procedure for Visual Assessment of Color Difference of Textiles, Option C (see 6.5), with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 (± 200) K.

4.6.2 Presence of labile sulfur test. In the determination of labile sulfur presence in textile materials with lead acetate, two (2) 1.50 (± 0.01) gram samples from each material submitted for evaluation shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. The samples shall be submersed in a stannous chloride solution that contains 100 grams of stannous chloride crystals ACS in 100 milliliters of hydrochloric acid ACS (35 percent concentration) and 50 milliliters of distilled water.

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A filter paper wet out with a 5.0 percent lead acetate solution shall be placed over the top of the test tube. The lead acetate solution contains 5.0 grams of lead acetate CP reagent grade and enough distilled water to make up a 100 milliliter solution; if the solution is not clear add a few drops (one at a time) of glacial acetic acid until the solution is clear. The test tube containing the textile sample, stannous chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as “pass” or “fail.

4.6.3 Toxicity test. When required, (see 6.2) an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If toxicity requirement (see 3.11) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Department or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The labels covered by this specification are intended for use in dress clothing items where they will be permanently visible and the use of another type label would adversely affect the appearance of the item.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type and size required (see 1.2)
c. Design and legend to be printed on labels (see 3.9)
d. Conformance inspection acceptance quality limits (AQLs) (see 4.3)
e. When toxicity testing is required (see 3.11, 4.6.3)
f. Inspection conditions (see 4.3.1)
g. Packaging (see 5.1)

6.3 Standard sample. For access to standard samples, address the contracting activity issuing the invitation for bids or request for proposals.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.6 Subject term (key word) listing.

Clothing
Dress
Polyester
Uniform

Custodians:
Army - GL
Navy – NU
Air Force -11

Preparing activity:
DLA-CT

(Project: 8315-2021-002)

Review activities
Army - MD
Navy - CG1

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